

Multichannel Histograms for Flow Classification

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Abstract—Encrypted traffic poses increasing challenges for effective network monitoring and management. To address this, machine learning and deep learning techniques are commonly applied to encrypted traffic classification by analyzing the statistical properties of network flows, such as packet lengths and inter-arrival times. Rather than relying on point estimates of these properties, we aim to obtain a comprehensive view of the distribution using histograms. In this paper, we propose and analyze a multichannel 2D histogram representation that incorporates packet lengths, directions, and inter-arrival times, extending the prior well-performing flowpic representation. We classify these histograms using a 2D convolutional neural network with residual connections, improving classification accuracy over previous methods.

Index Terms—Computer Network Monitoring, Traffic Classification, Neural Networks, Histograms

I. INTRODUCTION

The rapid expansion of global connectivity has placed increasing demands on the management and monitoring of modern networks. As encryption protocols such as encrypted DNS, TLS 1.3, and QUIC become more widely adopted, traditional techniques like deep packet inspection (DPI) are becoming less effective. This led to adoption of machine learning methods [1]–[3].

The choice of input representation is critical, as it directly affects the amount of information, its uniqueness, and the volume of data that must be stored or transferred. The input format also influences the model's achievable accuracy. In the context of flow traffic classification, machine learning methods typically rely on analyzing statistical properties of network flows. These properties include packet lengths, inter-arrival times, and packet directions, which provide insights into the behavior of network traffic.

Current methods aggregate statistical information to include only the first n packets of a flow [4], [5], or compute specially crafted features that summarize key aspects of the traffic [6], [7]. Commonly used features, such as the mean or standard deviation of packet lengths in different directions, may serve as point estimates of the distribution, capturing key characteristics of the traffic.

Histograms offer a different approach to representing network flows and can provide a more comprehensive view of the entire data distribution, particularly when dealing with complex traffic patterns, such as encrypted tunnels encapsulating

multiple services, as they allow the inference of underlying distributions from a broad range of families.

One promising format for representing network flow histograms is flowpic [8], a 2D histogram of packet lengths distributed over time. Flowpic was originally designed to leverage 2D convolutional neural network (CNN) models, adapted from the computer vision field. It captures the temporal evolution of packet lengths in a flow, transforming a series of packet lengths into a 2D image-like representation. Variations of flowpic [9] have been developed to reduce histogram sizes while maintaining effectiveness, and this reduced format has been adopted in several recent works [10]–[13].

In this paper, we investigate the current flowpic format and propose several enhancements to both the format and model. We evaluate the impact of integrating new information into multichannel histograms and classify them using a 2D CNN with residual connections, achieving improved classification accuracy over prior methods. Both the new proposed format and CNN design are directly applicable to research in flow traffic classification, data augmentation, and neural network processing.

II. RELATED WORK

The statistical representations of network flow data are widely used in traffic classification. Moore et al. [14] presented a comprehensive set of features for use in flow-based classification, all derived from previously mentioned packet statistics. These discriminators were shown to be applicable to encrypted traffic. A similar set of features has been applied to many tasks, such as the analysis of SSH [15], Tor [16], and malware [17].

Common approach is to limit the analysis to the first n packets of a flow. Lopez-Martin et al. [18] used the first 20 packets to classify traffic types, while Roy et al. [4] conducted experiments using sequences of 10, 20, and 30 packets. Bar et al. [19] focused on the first 100 packets, and Leroux et al. [20] collected the first 1024 packets.

Recently, there have been attempts to represent flows as images (2D matrices). Shapira et al. [8] introduced flowpic, a 2D histogram representation of packet lengths over time classified using a LeNet-5 architecture [21]. This format uses a 1500×1500 matrix that represents the flow's temporal and size characteristics. Later, the same authors [10] described

flowpic as a general approach for flow input representation, advocating for its use in various traffic classification tasks.

To address the storage and computational challenges posed by the large 1500×1500 matrix, Horowicz et al. [9] introduced mini-flowpic, which reduced the size to 32×32 and 64×64 matrices. This coarser binning still retained essential flow characteristics while being more practical for real-world applications. Finamore et al. [11] compared the original and mini-flowpic formats across multiple datasets, demonstrating the efficacy of the smaller representations. Okonkwo et al. [13] adapted the flowpic format to a 224×224 input size. Nevertheless, bigger matrices did not significantly improve accuracy and posed challenges due to the increased data volume. 32×32 is seemingly a good trade-off between histogram granularity and the amount of sent data and is, therefore, in the current research, the most common.

III. METHODOLOGY

In this study, the input data consists of metadata extracted from network packets, which are aggregated into flows, defined by a five-tuple: source IP, destination IP, source port, destination port, and transport protocol. We use bidirectional flows in form of histograms and employ and extend previously used formats. The following subsections formalize the original flowpic format and the enhancements we introduce.

A. Flowpic

The original flowpic may be calculated from the sequence of packet lengths l_i and their respective relative times t_i for the first n packets:

$$\begin{bmatrix} \mathbf{l} \\ \mathbf{t} \end{bmatrix} = \begin{bmatrix} l_1 & l_2 & \dots & l_n \\ t_1 & t_2 & \dots & t_n \end{bmatrix} \quad (1)$$

The packets are taken from the first 15 seconds of the flow ($t_n \leq 15$); though later work examined the use of a larger time threshold, we use the first 15 seconds for better comparability. Flowpic can then be defined as a 2D histogram of frequencies:

$$F = \{f_{ij}\} \quad (2)$$

where:

$$f_{ij} = \sum_{k=1}^n \mathbf{1}_{[(l_k, t_k) \in b_i^l \times b_j^t]} \quad (3)$$

for length bins $\{b_i^l\}$ and time bins $\{b_j^t\}$. The flowpic was calculated as 1500×1500 matrix with length bins for each individual length from 0 to MTU and time bins with a width of 0.01 seconds.

B. Packet Directions

Apart from packet lengths, we integrate packet directions d_i to differentiate packets from client to server and those sent from server to client, representing packet sequences as three vectors:

$$\begin{bmatrix} \mathbf{l} \\ \mathbf{d} \\ \mathbf{t} \end{bmatrix} = \begin{bmatrix} l_1 & l_2 & \dots & l_n \\ d_1 & d_2 & \dots & d_n \\ t_1 & t_2 & \dots & t_n \end{bmatrix} \quad (4)$$

Building on this concept, we propose to integrate packet directions into the flowpic format. Specifically, we chose to transform the original flowpic into two separate channels of length frequencies for each direction, designated as channels $F^{(1)}$ and $F^{(2)}$:

$$F^{(1)} = \{f_{ij}^{(1)}\} \quad (5)$$

$$F^{(2)} = \{f_{ij}^{(2)}\} \quad (6)$$

Note that the original single-channel definition can be retrieved by summing the two channels:

$$f_{ij} = f_{ij}^{(1)} + f_{ij}^{(2)} \quad (7)$$

C. Inter-arrival Times

Next, we include information about inter-arrival times. This temporal information can be expressed as the difference to the previous packet time:

$$\Delta t_i = t_i - t_{i-1} \quad (8)$$

Following Equation 4, input sequence can be formulated as:

$$\begin{bmatrix} \mathbf{l} \\ \mathbf{d} \\ \mathbf{t} \\ \Delta \mathbf{t} \end{bmatrix} = \begin{bmatrix} l_1 & l_2 & \dots & l_n \\ d_1 & d_2 & \dots & d_n \\ t_1 & t_2 & \dots & t_n \\ \Delta t_1 & \Delta t_2 & \dots & \Delta t_n \end{bmatrix} \quad (9)$$

We propose a four-channel implementation of flowpic to incorporate inter-arrival times. Channels $F^{(1)}$ and $F^{(2)}$ represent packet lengths for both directions, while channels $F^{(3)}$ and $F^{(4)}$ capture the inter-arrival times for both directions.

D. Binning

The original flowpic uses uniform bins with equal bin widths w_b , considering that both variables (relative time and length) have an implicit minimum (0) and maximum (15 seconds for relative time and MTU for packet length). Inter-arrival times, on the other hand, are heavily skewed. To address this, we decided to compute bins up to the 95th percentile to remove outliers that are far from most of the data. Figure 1 shows the distributions of packet lengths, relative times, and inter-arrival times we analyzed during the design process.

The distributions indicate that all three measured variables exhibit exponential characteristics. Thus, we also considered using bins with exponential widths to achieve more uniformly distributed frequencies in the histograms. Nevertheless, our prior experiments showed that this approach did not improve classification accuracy, so we maintained the standard uniform bin widths.

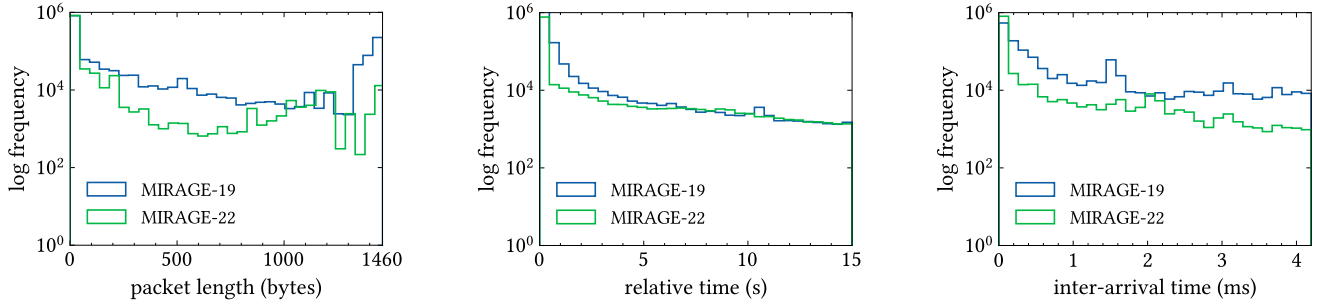


Fig. 1: Log-scaled frequency distributions of packet lengths, relative times, and inter-arrival times. All distributions are skewed and exhibit exponential characteristics. Packet lengths and relative times are shown from a minimum of 0 to a maximum of 1460 bytes and 15 seconds, respectively. Inter-arrival times are shown up to 4 milliseconds (95th percentile).

E. Model Architecture

We propose a model architecture inspired by the work of Wang et al. [22], which originally processed packet sequences using 1D CNN with residual building blocks [23]. Our proposed model employs a similar approach but incorporates several enhancements to improve performance and addresses identified limitations. The final model forms a network with two residual blocks, using 2D convolutional layers to process the multichannel flowpic input, followed by batch normalization. We identified that the pooling and the small size of the linear layer in the previous architecture were performance bottlenecks. Therefore, we adapted more fine-grained average pooling, and instead of a simple linear layer, the classification may utilize a final feed-forward network. The model architecture also includes dropout regularization. Full architecture is shown in Figure 2.

IV. EVALUATION

We evaluated the classification accuracy of formats described in the previous sections (single-channel, two-channel, and four-channel) together with representation of raw packet series. In addition to presenting the main results and accuracies, we examine the theoretical upper accuracy limit of the formats due to granularity of the input. All code for data preparation, model training, and experiments is available¹.

A. Datasets

We selected two public datasets that have been used in previously published papers related to flowpic and encrypted traffic classification: MIRAGE-19 [24] and MIRAGE-22 [25]. Both of these datasets capture user traffic from mobile applications, along with traffic logs and system calls, to ensure reliable ground truth labels linking the traffic to the source applications, such as Facebook, Dropbox, Spotify, Twitter, Slack, or Skype. Moreover, these two datasets target different types of traffic. The MIRAGE-19 dataset predominantly contains short flows with relatively few packets per flow, while the MIRAGE-22 dataset, focused on online calls and videoconferencing, consists primarily of long-term flows with

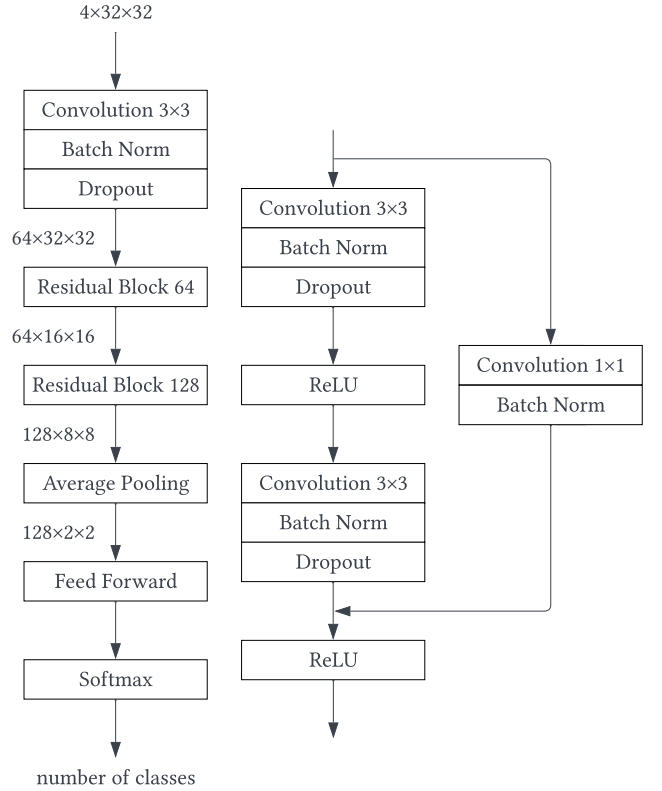


Fig. 2: Proposed model architecture with residual blocks, following and improving [22].

many packets—having a mean of 17 non-zero packets per flow in MIRAGE-19 and 6,598 in MIRAGE-22.

To ensure consistency with previous studies, we used the same preprocessing techniques and verified our results with published statistics [22]. This included ensuring the datasets had exactly the same number of flows, the same number of labels, and a similar distribution of packets per flow. Specific preprocessing steps included ensuring a minimum of 10 packets in each flow, or removing zero-length packets (such as TCP SYN, ACK, and FIN).

¹<https://github.com/danielpoliakov/flowmind>

TABLE I: Comparison of classification accuracy (%) for single, two, and four-channel flowpic formats.

Method	MIRAGE-19	MIRAGE-22
32×32 (single-channel)	77.02	95.36
32×32 (two-channel)	80.21	95.96
32×32 (four-channel)	81.10	96.71
16×16 (four-channel)	78.95	95.79

B. Experimental Setup

During the experiments, datasets were divided into training, validation, and testing sets using an 80:10:10 split. The final model architecture, shown in Figure 2, includes a dropout rate of 0.2 and a feed-forward classifier with layers structured as follows: a fully connected layer from 512 to 256 units, followed by a ReLU activation, and another fully connected layer from 256 units to the number of output classes. We also evaluated different hyperparameters, such as smaller dropout rates and a linear classifier instead of a feed-forward network. These alternative models exhibited the same trend as seen in the presented results. All models were trained for 50 epochs using AdamW optimizer with cosine annealing learning rate scheduling. The initial learning rate is 0.001, and the weight decay is 0.0001. The primary metric for evaluation was accuracy.

C. Results

The results in Table I demonstrate that incorporating additional channels into the flowpic format leads to improvements in classification accuracy. The single-channel implementation achieved an accuracy which already outperformed the baseline from Finamore et al. [11] due to our model update. The baseline accuracies for the 32×32 flowpic were 69.91 % and 90.97 % for the two datasets, following the methodology of Horowicz et al. [9] in a replication study [11] using a LeNet-5 based model. The study used the same dataset, preprocessing steps, and splits. Using their model, we were able to reproduce the mentioned results. Adding packet directions (two-channel flowpic) further boosted accuracy on both datasets. Finally, incorporating both packet directions and inter-arrival times in the four-channel flowpic yielded the highest accuracy, reaching 81.10 % on MIRAGE-19 and 96.71 % on MIRAGE-22.

Additionally, we evaluated the accuracy of 16×16 flowpic. With four channels, this format contains the same amount of data as the original 32×32 , since $32 \times 32 = 4 \times 16 \times 16$. Our results show that while reducing the size to 16×16 slightly decreases accuracy, it still outperforms the single-channel 32×32 flowpic. This suggests that our approach can maintain high classification accuracy without negatively impacting flow transfer and storage efficiency.

In Table II, we compare the performance of our best four-channel flowpic model against packet series and the 32×32 baseline flowpic from Finamore et al. [11]. Our four-channel model outperformed both, achieving a 5.67 % improvement on MIRAGE-19 over the baseline flowpic and a 1.79 % improvement on MIRAGE-22. Similarly, our method showed

TABLE II: Comparison of classification accuracy (%) between our four-channel representation, packet series [22], and baseline histograms from prior work [11].

Method	MIRAGE-19	MIRAGE-22
Wang et al. [22]	75.43	94.92
Finamore et al. [11]	69.91	90.97
ours	81.10	96.71

a significant improvement over the packet series approach by Wang et al. [22], further demonstrating the effectiveness of the proposed histograms.

The overall results indicate that incorporating additional channels into the flowpic format significantly improves classification accuracy. The four-channel implementation shows the highest performance, demonstrating the value of including packet lengths and inter-arrival times for both directions.

D. Accuracy Limit

During the evaluation, we observed that due to the nature of histograms, some flowpic representations overlapped, meaning two flowpic matrices of packet lengths contained the same values. Moreover, overlapping data sometimes belonged to different classes. Hence, the maximum achievable accuracy of the models is inherently less than 100 %.

The maximum theoretical accuracy can be expressed for a multi-class classification problem using an ideal model. Let the true label be y_i , and the prediction of the model be $\hat{y}_i$. An ideal model knows the true label and predicts $\hat{y}_i = y_i$ if possible. However, since some inputs have collisions and for the same input, there are multiple true labels y_i , the ideal model chooses the majority label for the colliding input.

Our proposed enhancements lessen these limitations. For MIRAGE-19 dataset, the upper accuracy limit is 96.99 % with the standard single-channel flowpic, and it improves to 99.91 % with the four-channel implementation. Similarly, for MIRAGE-22, the upper accuracy limit is 99.08 %, which improves to 99.81 % with the four-channel implementation.

V. CONCLUSION

In this paper, we analyzed the flowpic format and proposed several enhancements to improve its classification accuracy in the context of encrypted network traffic. We formalized earlier histogram representation and introduced a multichannel format that incorporates additional information. This format was evaluated using a custom CNN model that outperformed existing methods. These advancements can be directly applied to the mentioned prior work, currently using input formats for representation learning and augmentations experiments. Additionally, investigating the impact of different types of network traffic and encryption protocols on the histogram-based methods could provide further deeper insights.

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